# Introduction to Probability and Statistics

Lecture 6: Some Classical Continuous distributions

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## Uniform distribution

### Definition

*A r.v. $X$ follows a uniform distribution on the domain $[a, b] \subset \mathbb{R}$ iff its p.d.f. is of the form:*

$$f(x) = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x)$$

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*A r.v. $X$ has an exponential distribution with parameter $\theta > 0$ iff its p.d.f. has the form:*

$$f(x \mid \theta) = \frac{1}{\theta} \exp\left\{-\frac{x}{\theta}\right\}, \; x \in \mathbb{R}^+.$$

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- Cumulative distribution function: $F(x) = 1 - e^{-x/\theta}$.
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- Expectation: $E(X) = \theta$.
- Variance: $Var(X) = \theta^2$.

## Gamma distribution

### Definition

*A r.v. $X$ has a gamma distribution with parameter $\alpha, \beta > 0$ iff its p.d.f. has the form:*

$$f(x \mid \alpha, \beta) = \frac{\beta^{\alpha}}{\Gamma(\alpha)}\, x^{\alpha-1} e^{-\beta x}, \; x \in \mathbb{R}^{+}$$

*where $\Gamma(.)$ is the gamma function defined by:* $\Gamma(\alpha) = \int_0^{+\infty} u^{\alpha-1}\, e^{-u}\, du.$

- Notation: $X \sim \mathscr{G}(\alpha, \beta)$
- Expectation: $E(X) = \alpha/\beta$.
- Variance: $Var(X) = \alpha/\beta^2$.
- Moment generating function: $M_X(t) = \left(\frac{\beta}{\beta - t}\right)^{\alpha}$.

## Gamma distribution

**Exercise**

Let $X_1, X_2, \ldots, X_n$ be $n$ independent exponential r.v with parameter $\theta$.

Find the distribution of $S_n = \sum_{i=1}^{n} X_i$ using the moment generating function.

# Normal (Gaussian) distribution

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### Definition

*A r.v. $X$ has a normal with parameters $(\mu, \sigma^2)$ $\mu \geqslant 0$, $\sigma^2 > 0$ iff its p.d.f. is:*

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, \quad x \in \mathbb{R}.$$

- Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$.

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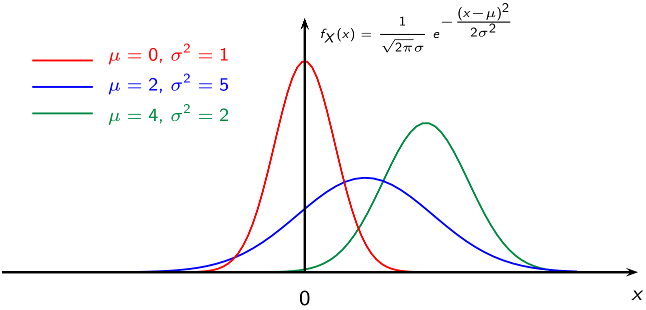
- Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$.
- Cumulative distribution function:

$$F(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma}} \exp\left\{-\frac{(t-\mu)^2}{2\sigma^2}\right\} dt.$$

# Normal (Gaussian) distribution

*p.d.f. of a normal r.v for differents parameters values*



# Normal (Gaussian) distribution

## Normal distribution

- When $\mu = 0$ and $\sigma^2 = 1$, $X$ is said to follow the **standard normal distribution**.
- p.d.f. of the standard normal distribution:

$$f(x \mid 0, 1) = \phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}, \quad x \in \mathbb{R}.$$

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## Normal distribution

Checking that $\varphi$ is a pdf that is $\displaystyle\int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}}\ e^{-\frac{z^2}{2}}\ dz = 1$.

Let $I = \int_{\mathbb{R}} e^{-\frac{z^2}{2}}\ dz$ and compute $I^2$.

$$I^2 = \left(\int_{\mathbb{R}} e^{-\frac{u^2}{2}}\ du\right)\left(\int_{\mathbb{R}} e^{-\frac{v^2}{2}}\ dv\right) = \int_{\mathbb{R}}\int_{\mathbb{R}} e^{-\frac{u^2+v^2}{2}}\ dudv.$$

Making the change of variables $u = r\cos\theta$ and $v = r\sin\theta$, we have $du\ dv = rdrd\theta$. The integration domain becomes $[0,+\infty[\times[0,2\pi]$.

$$\begin{aligned} I^2 &= \int_0^{+\infty}\int_0^{2\pi} e^{-r^2/2} rdrd\theta = \left(\int_0^{2\pi} d\theta\right)\times\left(\int_0^{2\pi}\ r\ e^{-r^2/2}dr\right) \\ &= \left[-e^{-r^2/2}\right]_0^{+\infty} = 2\pi. \end{aligned}$$

Therefore $I = \sqrt{2\pi}$.

## Normal distribution

### Proposition

$$\left(X \sim \mathcal{N}(\mu, \sigma^2)\right) \iff \left(Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1)\right).$$

**Proof:** Express $P(X \leqslant x)$.Then $P(Z \leqslant z)$.

## Normal distribution

**Proof:** Suppose $X \sim \mathcal{N}(\mu, \sigma^2)$. What is the distribution of $Z = (X-\mu)/\sigma$?

$$P(Z \leqslant z) = P\left(\frac{X-\mu}{\sigma} \leqslant z\right) = P(X \leqslant \mu+\sigma z) = \int_{-\infty}^{\mu+\sigma z} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(t-\mu)^2}{2\sigma^2}\right\} dt.$$

Let's set $u = (t-\mu)/\sigma$. Then $t = \sigma u + \mu$, $dt = \sigma du$ and when $t$ varies between $-\infty$ and $\mu + \sigma z$, $u$ varies between $-\infty$ and $z$.
Thus

$$P(Z \leqslant z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{u^2}{2}\right\} du = \Phi(z).$$

Conversely, consider $Z \sim \mathcal{N}(0,1)$. What is the distribution of $X$ such that $Z = (X-\mu)/\sigma$ i.e. the distribution of $X = \sigma Z + \mu$.

$$P(X \leqslant x) = P(\sigma Z + \mu \leqslant x) = P\left(Z \leqslant \frac{x-\mu}{\sigma}\right) = \int_{-\infty}^{(x-\mu)/\sigma} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{u^2}{2}\right\} du.$$

Let's set $u = (t-\mu)/\sigma$.The $du = dt/\sigma$ and when $u$ is in $]-\infty, (x-\mu)/\sigma]$, $t$ is $]-\infty, x]$.
Thus

$$P(X \leqslant x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{(t-\mu)^2}{2\sigma^2}\right\} \frac{dt}{\sigma}.$$

We recognize the normale distribution with parameters $(\mu, \sigma^2)$.

## Normal distribution

**Exercise**

Compute the moment generating function (m.g.f.) of a standard normal r.v.

Find the m.g.f of a normal r.v. with parameters $(\mu, \sigma^2)$ using the previous proposition.

## Normal distribution

**Exercise**
Compute the moment generating function (m.g.f.) of a standard normal r.v.
Find the m.g.f of a normal r.v. with parameters $(\mu, \sigma^2)$ using the previous proposition.

$$\left(X \sim \mathcal{N}(\mu, \sigma^2)\right) \Longleftrightarrow \left(M_X(t) = \exp\left\{\mu t + \frac{\sigma^2}{2} t^2\right\}\right).$$

## Normal distribution

- **Expectation**

  Let $Z \sim \mathcal{N}(0,1)$, then $E(Z) = 0$.

  Let $X = \mu + \sigma Z$, $X \sim \mathcal{N}(\mu, \sigma^2)$ and $E(X) = \mu$.

- **Variance**

  $Var(Z) = E(Z^2) - (EZ)^2$.

  It can be shown that: $E(Z^2) = 1$. Thus: $Var(Z) = 1$.

  $Var(X) = Var(\mu + \sigma Z) = \sigma^2 Var(Z) = \sigma^2$.

- **Conclusion**

  If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $E(X) = \mu$ and $Var(X) = \sigma^2$.

## Normal distribution

### Proposition

$$\forall z, \ \Phi(z) + \Phi(-z) = 1.$$

**Proof:** compute $P(Z \leqslant -z)$.

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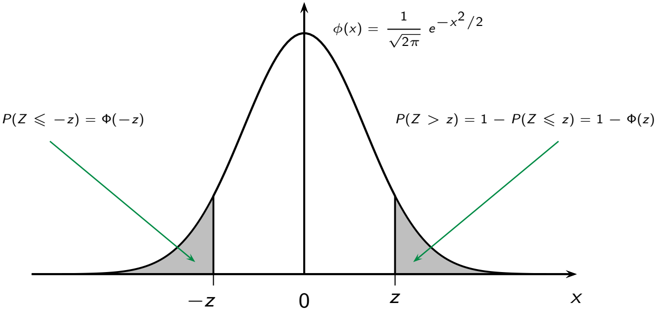
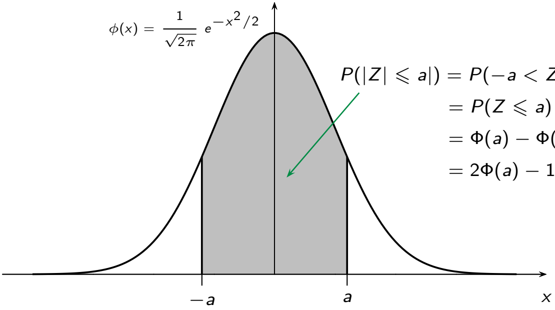


## Normal distribution

### Proposition

Soit $Z \sim \mathcal{N}(0,1)$, $P(|Z| \leqslant a) = 2\Phi(a) - 1$.




$$\begin{aligned} P(|Z| \leqslant a|) &= P(-a < Z \leqslant a) \\ &= P(Z \leqslant a) - P(Z \leqslant -a) \\ &= \Phi(a) - \Phi(-a) \\ &= 2\Phi(a) - 1 \end{aligned}$$

## Normal distribution

Remark: $P(|Z| > a) = 1 - P(|Z| \leqslant a) = 1 - 2\Phi(a) + 1 = 2(1 - \Phi(a))$.

Let $P(|Z| \leqslant a) = \gamma$, then $\Phi(a) = \dfrac{1+\gamma}{2}$.

The quantile $a$ associated with the probability $\gamma$ is $a = \Phi^{-1}\left(\dfrac{1+\gamma}{2}\right)$.

## Normal distribution

Let consider $X \sim \mathcal{N}(\mu, \sigma^2)$.
We have:

$$P\left(\left|\frac{X-\mu}{\sigma}\right| \leqslant a\right) = \gamma \iff P(\mu - a\sigma < X \leqslant \mu + a\sigma) = \gamma.$$

Therefore the probability that the r.v. lies in an interval of length $2a\sigma$ centered on $\mu$ is equal to $\gamma$.

- For $a = 1$, $\Phi(1) = 0.8413$ and $\gamma = 2 \times 0.8413 - 1 = 0.6826$.
- For $a = 2$, $\Phi(2) = 0.9772$ and $\gamma = 2 \times 0.9772 - 1 = 0.9544$.
- For $a = 3$, $\Phi(3) = 0.9986$ and $\gamma = 2 \times 0.9986 - 1 = 0.9973$.